

Math 115A Spring 2026 Discussion 15 Handout

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1 Introduction

Today we'll review problems from the midterm, and discuss the algebraic and geometric multiplicity of eigenvalues.

2 Midterm Review

2.1 True or False?

1. Let V and W be finite-dimensional vector spaces such that $\dim(V) = \dim(W)$. If $T : V \rightarrow W$ is injective, then it is bijective.
2. Let V be a finite-dimensional vector space and $T : V \rightarrow V$ linear. Then $\operatorname{im}(T) + \ker(T) = V$.
3. Let V be a finite-dimensional vector space and $T : V \rightarrow V$ linear. Then $\det(2T) = 2 \det(T)$.
4. Let V be a vector space and $T : V \rightarrow V$ linear. If $x \in \ker(T)$, then x is an eigenvector of T .
5. Let V be a finite-dimensional vector space and $T : V \rightarrow V$ linear. If T^2 is diagonalizable, then T is diagonalizable.
1. **True:** Since T is injective, $\dim(\ker(T)) = 0$. Then $\dim(\operatorname{im}(T)) = \dim(V) = \dim(W)$, so T is surjective; thus, T is both injective and surjective, which implies it is bijective.
2. **False:** If you're not looking closely, this seems convincing - it looks almost like the Dimension Theorem. What should set off alarm bells is that, for a general map $T : V \rightarrow W$ (where $V \neq W$), $\operatorname{im}(T)$ and $\ker(T)$ do not live in the same vector space, and in such cases the sum does not even make sense. In this case, the subspace addition is defined, since $\operatorname{im}(T)$ and $\ker(T)$ are both subspaces of V , but we should feel that this operation is "unnatural" and therefore unlikely to have such a nice property. A counterexample is the map $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ given (in the standard basis) by the matrix $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$. Here $\operatorname{im}(T) = \ker(T) = \operatorname{Span}(\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix} \})$.

3. **False:** The intuition is that determinant scales with the size of a matrix. That is, if V is n -dimensional and $T \in \mathcal{L}(V)$, we have $\det(\lambda T) = \lambda^n \det(T)$. The easiest counterexample is the identity on $\mathbb{R}^2$: $\det(I) = 1$ but $\det(2I) = 4$.
4. **False:** This one is kind of mean, but remember that an eigenvector must be nonzero, whereas $0 \in \ker(T)$ for every linear map T .
5. **False:** This is another subtle one. As an example, let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be given (with respect to the standard basis) by $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$. We can check that the characteristic polynomial of T is x^2 , so its only eigenvalue is 0; if it were diagonalizable, it would be the zero matrix, which it isn't. But you can check that $T^2 = 0$, so it is diagonalizable.

2.2 Problem 2

Let V be a finite-dimensional vector space over a field F and $S, T \in \mathcal{L}(V)$. Suppose $S \circ T = I_V$. Prove that $T \circ S = I_V$.

Proof. This problem is a bit hard to find a foothold in, but once you make the right opening move, there are quite a few ways to do it. The key observation is:

Lemma: Let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ be functions (of any kind) and suppose $g \circ f : X \rightarrow Z$ is bijective. Then g is surjective and f is injective. [More generally: if $g \circ f$ is surjective, then g is surjective; if $g \circ f$ is injective, then f is injective].

You might not have guessed this statement, but once you know it's true, it's a one-line proof. (Try it!)

We know that $S, T : V \rightarrow V$ are linear maps and that $S \circ T$ is the identity. It follows that S is surjective and T is injective; since the domain of S has the same dimension as the codomain of S , and likewise for T , we know that in fact S and T are both isomorphisms.

Now the fun can begin. For example, observe that, because $S \circ T = Id$, $T \circ S \circ T = T$ (apply T on the left on both sides). So, for any $v \in V$, $(T \circ S)(Tv) = Tv$. We want to show that for any $w \in V$, $(T \circ S)w = w$. Since T is surjective, we can find $v \in V$ such that $Tv = w$. Then $(T \circ S)w = (T \circ S)Tv = Tv = w$.

Another approach is to use the fact that if S is invertible, then it has an inverse S^{-1} . We then have to show that "left inverses are unique;" this is essentially the same idea as the proof that inverses in a field are unique. We write:

$$\begin{aligned} S \circ S^{-1} &= S \circ T \\ S^{-1} \circ S \circ S^{-1} &= S^{-1} \circ S \circ T \\ Id_V \circ S^{-1} &= Id_V \circ T \\ S^{-1} &= T \end{aligned}$$

Since $S^{-1} \circ S = Id$, it follows that $T \circ S = Id$. □

2.3 Problem 3

Let V be a finite dimensional vector space over a field F , and let $T : V \rightarrow V$ be a linear map such that $T^2 = T$.

- (a) Show that $\sigma(T) \subseteq \{0, 1\}$
 - (b) Show that $\text{im}(T) = \ker(\text{Id}_V - T)$
 - (c) Prove that T is diagonalizable.
- (a) *Proof.* Consider an eigenvector v of T , so that $Tv = \lambda v$. Then $T^2v = \lambda^2v$; on the other hand, $Tv = T^2v$ by hypothesis, so we have $\lambda v = \lambda^2v$, hence $\lambda = \lambda^2$. The only solutions are $\lambda = 0$ or $\lambda = 1$. $\square$
- (b) *Proof.* We prove two inclusions. First, suppose $v \in \text{im}(T)$. Then $v = Tw$ for some $w \in V$. Now, consider $(\text{Id}_V - T)v$. This is equal to

$$v - Tv = v - T(Tw) = v - T^2w = v - Tw = v - v = 0$$

so $v \in \ker(\text{Id}_V - T)$.

Now, suppose $v \in \ker(\text{Id}_V - T)$. Then $v = Tv$, so $v \in \text{im}(T)$. $\square$

- (c) *Proof.* This solution might be a bit hard to spot: we've just seen in a true or false question that $\ker(T) + \text{im}(T)$ need not equal V , but in this case, equality holds. Let's see why.

Observe that T is the identity on $\text{im}(T)$: for $v \in \text{im}(T)$, we can write $v = Tw$ for some w , and we see that $Tv = T^2w = Tw = v$. Also, observe that $\dim(\text{im}(T)) + \dim(\ker(T)) = \dim(V)$, by the dimension theorem. Let $\{e_1, \dots, e_p\}$ be a basis for $\text{im}(T)$ and $\{k_1, \dots, k_q\}$ be a basis for $\ker(T)$. Observe that $Te_i = e_i$ and $Tk_j = 0$, so the e_i 's and k_j 's are eigenvectors of T . I claim that $\{e_1, \dots, e_p, k_1, \dots, k_q\}$ is a basis for V . Certainly, it has the right number of elements; if we can show that it is linearly independent, we will be done.

Suppose $\sum_i a_i e_i + \sum_j b_j k_j = 0$; we want to show that $a_i = b_j = 0$ for all i, j . Apply T to both sides of the equation; we get

$$\begin{aligned} \sum_i a_i T e_i + \sum_j b_j T k_j &= T(0) \\ \sum_i a_i e_i &= 0 \end{aligned}$$

(We know that $Tk_j = 0$ because $k_j \in \ker(T)$). Since the e_i 's form a basis for $\text{im}(T)$, they are linearly independent, so we must have $a_i = 0$ for all i . Then our linear combination reduces to

$$\sum_j b_j k_j = 0$$

But the k_j 's form a basis for $\ker(T)$; they are linearly independent, so we must have $b_j = 0$ for all j . So, our linear combination $\sum a_i e_i + \sum b_j k_j = 0$ must be trivial. We conclude that $\{e_1, \dots, e_p, k_1, \dots, k_q\}$ is a linearly independent subset of V of size $\dim(V)$; therefore, it is a basis. Since all the vectors in it are eigenvectors, it is an eigenbasis, and we conclude that T is diagonalizable. $\square$

If you remember some facts about multiplicity, you can prove this much more efficiently. (This is Sihao's official solution):

Proof. We want to show that $\sum_{\lambda \in \sigma(T)} \mathfrak{g}_\lambda = \dim(V)$. There are only two eigenvalues, 0 and 1; $\mathfrak{g}_0 = \dim(\ker(T))$ and $\mathfrak{g}_1 = \dim(\ker(Id_V - T)) = \dim(\operatorname{im}(T))$ (by part (b)). But $\dim(\ker(T)) + \dim(\operatorname{im}(T)) = \dim(V)$ for any finite-dimensional vector space V , by the Dimension/Rank-Nullity Theorem. $\square$

3 Algebraic and Geometric Multiplicity

First, as a warm-up here's a computational problem about ordinary eigenvalues (adapted from *Linear Algebra Done Wrong*) which might be enlightening:

Exercise: Construct a matrix A with eigenvalues 1 and 3 corresponding to eigenvectors $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

Now, on to multiplicity:

We have, essentially, two equivalent definitions of eigenvalue:

Definition: A *geometric* eigenvalue for T is a scalar λ such that for some nonzero $v \in V$, $Tv = \lambda v$.

Definition: An *algebraic* eigenvalue for T is a scalar λ such that $\det(\lambda I - T) = 0$, i.e. a solution to the characteristic polynomial χ_T of T .

Now, there are obvious ways for each type of eigenvalue to "repeat."

Definition: The multiplicity of a geometric eigenvalue λ of T , more commonly called its *geometric multiplicity* denoted $\mathfrak{g}_\lambda$, is the dimension of the eigenspace $E_\lambda = \{v \in V : Tv = \lambda v\}$; you can also think of this as $\dim(\ker(\lambda I - T))$.

Definition: The multiplicity of an algebraic eigenvalue λ of T , more commonly called its *algebraic multiplicity* and denoted $\mathfrak{a}_\lambda$, is the multiplicity of the root λ in the characteristic polynomial χ_T . That is, if $\chi_T(x)$ is the characteristic polynomial of x , $\mathfrak{a}_\lambda$ is the largest n such that the polynomial $(x - \lambda)^n$ divides $\chi_T(x)$.

Easy exercise: Let T be given by the matrix

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 2 \\ 0 & 1 & 0 & 0 & 3 \\ 0 & 0 & 5 & 2 & 5 \\ 0 & 0 & 0 & 5 & 800 \\ 0 & 0 & 0 & 0 & 5 \end{pmatrix}$$

Find $\sigma(T)$. For each $\lambda \in \sigma(T)$, find $\mathbf{a}_\lambda$.

More difficult exercise: With T as above, find $\mathbf{g}_\lambda$ for each $\lambda \in \sigma(T)$. (Hint: work one value of λ at a time. You want to compute $\dim \ker(\lambda I - T)$; using the Dimension theorem, you can rephrase this as an easier problem about the rank of T).

The multiplicity of eigenvalues is another “invariant of linear maps.” That is, if $T : V \rightarrow V$ is a linear map and $S = P^{-1}TP$ for some invertible $P : V \rightarrow V$, we know not only that $\sigma(T) = \sigma(S)$, but also that for each $\lambda \in \sigma(T) = \sigma(S)$, $\mathbf{a}_\lambda(T) = \mathbf{a}_\lambda(S)$ and $\mathbf{g}_\lambda(T) = \mathbf{g}_\lambda(S)$. (The second statement is easy to check directly; the first follows from the fact that $\chi_S = \chi_T$, which follows from properties of the determinant).

It should be clear how to interpret $\mathbf{g}_\lambda$ in terms of the multiplicity of T : namely, it tells you the size of an “eigenspace” for λ , a subspace of V which T preserves and scales up without rotating. You may be wondering whether $\mathbf{a}_\lambda$ has any similar interpretation. You have already seen that $\mathbf{g}_\lambda \leq \mathbf{a}_\lambda$, but in fact the algebraic multiplicity contains more detailed information:

Exercise: Consider the matrix $M = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, which we can view as a linear map $\mathbb{R}^2 \rightarrow \mathbb{R}^2$.

1. Compute χ_M and $\sigma(M)$.
2. Notice that $\sigma(M)$ consists of a single eigenvalue λ . Compute $\mathbf{g}_\lambda$ and $\mathbf{a}_\lambda$.
3. Suppose $v \in \ker((I - M)(I - M))$. What must be true of $(I - M)v$?
4. Compute $(I - M)^2$.
5. Repeat but with $\begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$. What do you notice?

If you go on to take Math 115B, you’ll learn about the Jordan normal form, which provides, at last, a complete characterization of all linear maps $V \rightarrow V$; in particular, it is the closest thing to “diagonalization” that can be applied to all square matrices. This geometric interpretation of algebraic multiplicity will play a central role in understanding this Normal Form.